

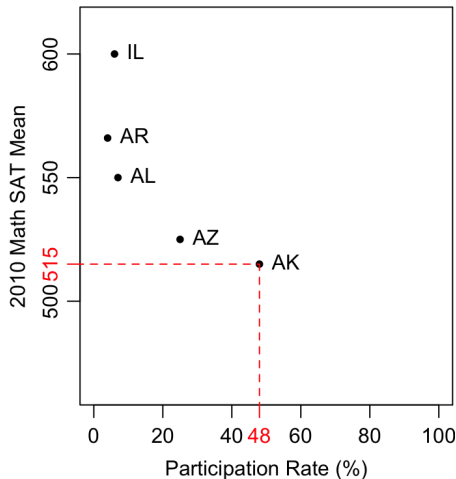
STAT 20000 Chapter 8

Scatterplots & Correlation

How to Draw A Scatter Diagram

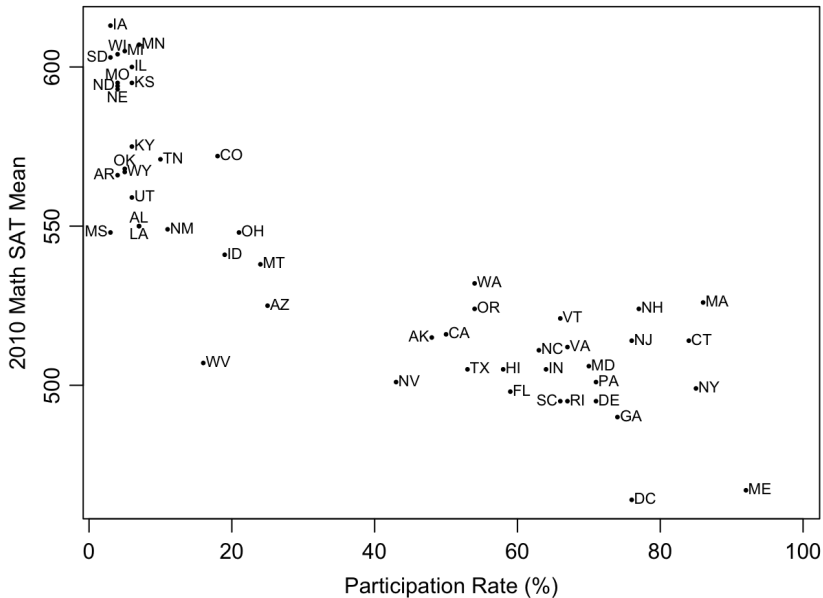
2010 Mean Math SAT Scores¹

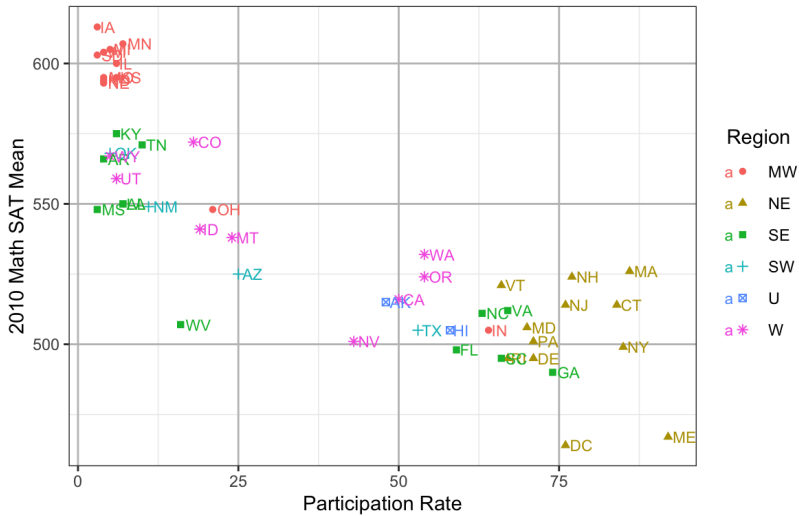
State	Participation Rate ²	Math
AK	48	515
AL	7	550
AR	4	566
AZ	25	525
CA	50	516
⋮		
IL	6	600
⋮		
WY	5	567



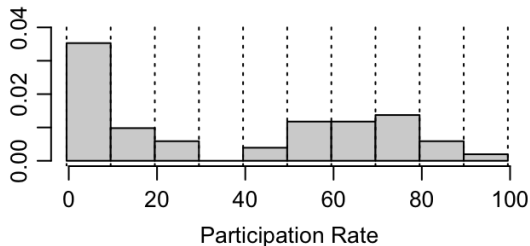
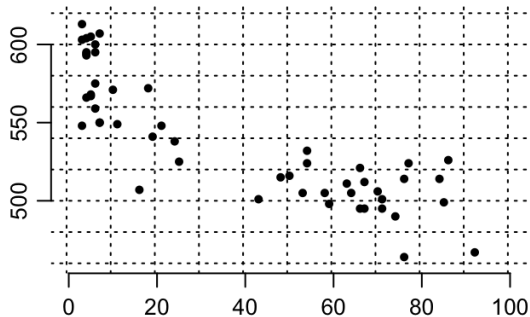
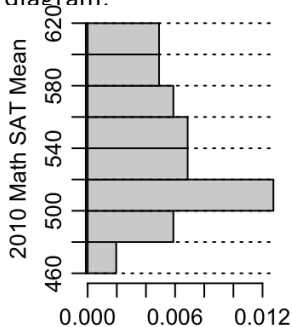
¹Source: CollegeBoard

²The percentage of high school graduates in 2010 that took SAT

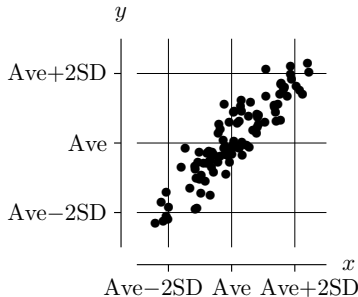




One can get the histograms of individual variables from a scatter diagram.



Summarizing a Scatter Diagram



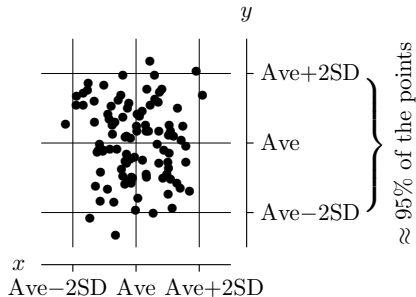
$\approx 95\%$ of the points

Strong association

Tight clustering

Knowing x helps
in predicting y

Correlation near 1



$\approx 95\%$ of the points

Weak association

Loose clustering

Knowing x isn't much
help in predicting y

Correlation near 0

Summarizing a Scatter Diagram (Cont'd)

A histogram can be summarized by

- ▶ average — center
- ▶ SD — spread

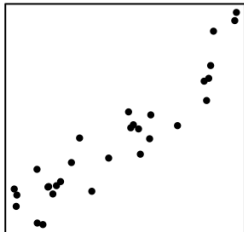
A scatter diagram can be summarized by 5 statistics:

- ▶ The average and SD of the x -values
- ▶ The average and SD of the y -values
- ▶ The correlation coefficient r

The averages and the SDs specify the center and spread of the cloud of points, horizontally and vertically. The r measures the amount of linear association, i.e., clustering about a line

Forms of Relationship

Linear



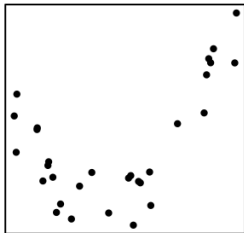
Linear



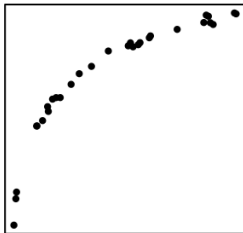
No Relation



Nonlinear



Nonlinear



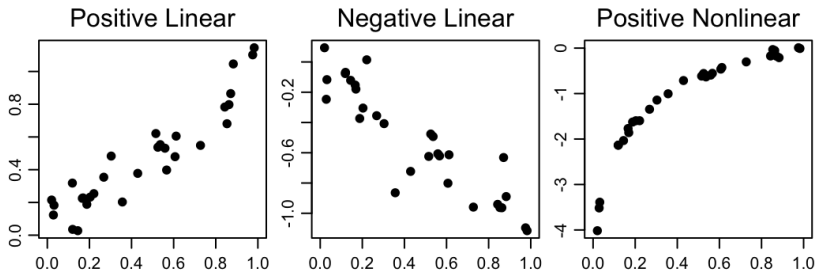
No Relation



Positive and Negative Associations

Positive association: High values of one variable tend to occur together with high values of the other variable.

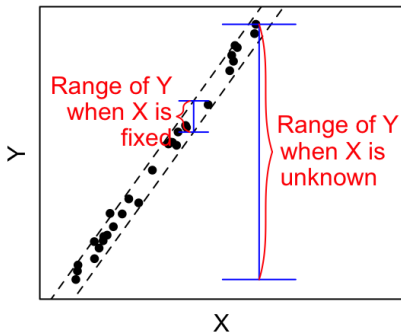
Negative association: High values of one variable tend to occur together with low values of the other variable.



Strength of Association

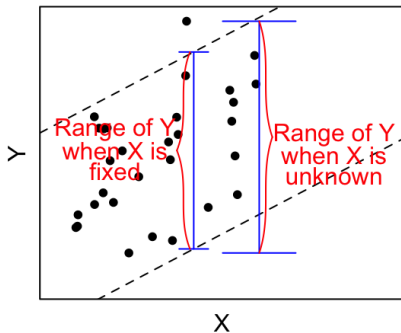
The **strength** of the relationship between the two variables can be seen by how much variation, or **scatter**, there is around the main form.

Strong Association



Small spread of Y
when X is known

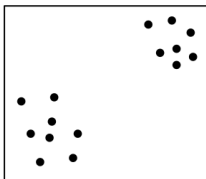
Weak Association



Large spread of Y
when X is known

Clusters & Outliers

Sometimes points in a scatter plot form *clusters*, which indicates data may be comprised of several distinct kinds of individuals.



Scatter plots can also be used to spot *outliers*.



Recap: What to Look For in a Scatterplot?

- ▶ What is the *form* of the relationship?
(linear, curved, clustered ...)

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Recap: What to Look For in a Scatterplot?

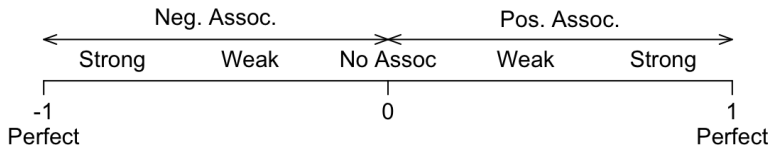
- ▶ What is the *form* of the relationship?
(linear, curved, clustered ...)
- ▶ What is the *direction* of the relationship?
(positive, negative)
- ▶ What is the *strength* of the relationship?
(strong, weak, ...)
- ▶ Are the points *clustered*?
- ▶ Are there any deviations from the overall pattern, i.e. *outliers*?
- ▶ In fact, the list above is not exclusive.
Any unusual pattern worth investigation.

Correlation = Correlation Coefficient, r

Correlation r is a numerical measure of the **direction** and **strength** of the **linear** relationship between two numerical variables.

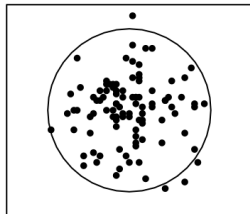
“ r ” always lies between -1 and 1 ; the strength increases as you move away from 0 to either -1 or 1 .

- ▶ $r > 0$: positive association
- ▶ $r < 0$: negative association
- ▶ $r \approx 0$: very weak linear relationship
- ▶ large $|r|$: strong linear relationship
- ▶ $r = -1$ or $r = 1$: *only* when all the data points on the scatterplot lie exactly along a **straight line**

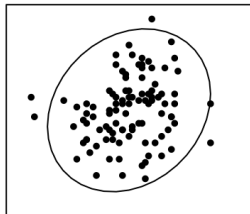


Positive Correlations

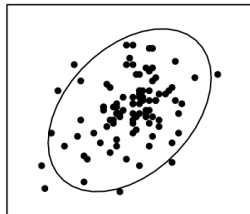
$r = 0$



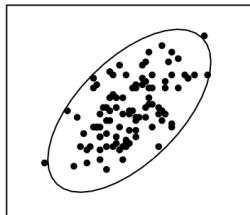
$r = 0.2$



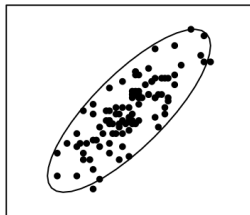
$r = 0.4$



$r = 0.6$



$r = 0.8$

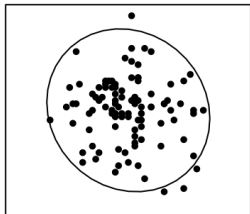


$r = 0.9$

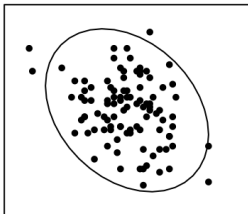


Negative Correlations

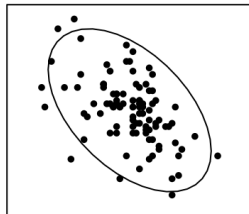
$r = -0.1$



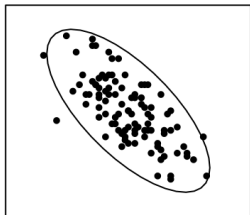
$r = -0.3$



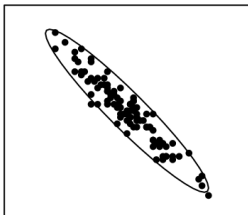
$r = -0.5$



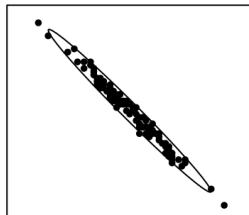
$r = -0.7$



$r = -0.95$



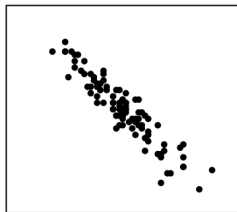
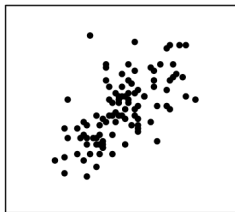
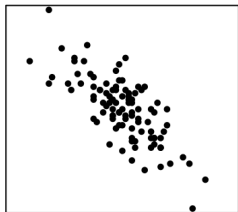
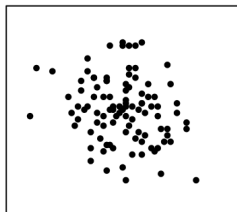
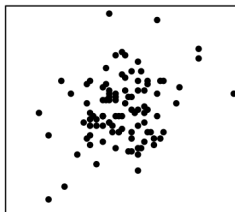
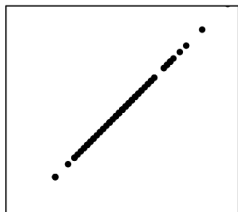
$r = -0.99$



Exercises: Guess r by Eye

Match the scatterplots below with the correlation coefficients.

-0.93 , -0.75 , -0.20 , 0.27 , 0.63 , and 1.0

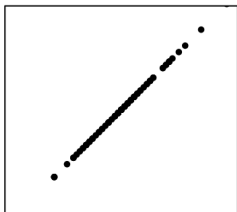


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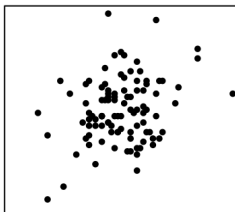
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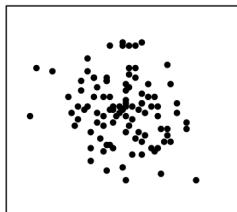
$r = 1$



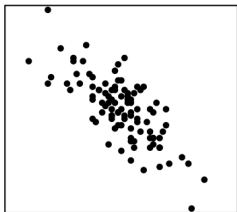
$r = 0.27$



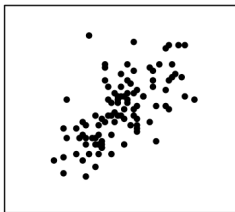
$r = -0.2$



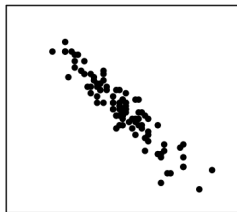
$r = -0.75$



$r = 0.63$



$r = -0.93$



Computation of the Correlation Coefficient (r)

$$\begin{array}{l} (x_1, y_1) \\ (x_2, y_2) \\ (x_3, y_3) \end{array} \quad r = \frac{1}{n} \sum_i \left(\frac{x_i - \bar{x}}{SD_x} \right) \left(\frac{y_i - \bar{y}}{SD_y} \right)$$

= Ave. of (x in standard units) $\times$ (y in standard units).

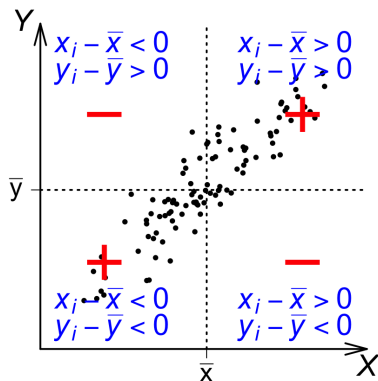
$\vdots$

That is

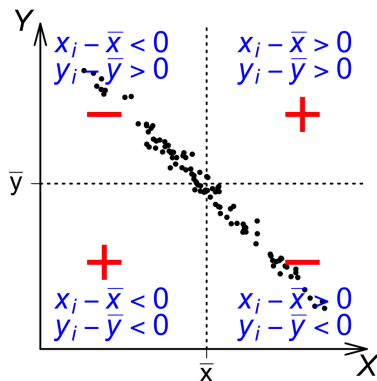
- (x_n, y_n)
- ▶ convert each x-value and y-value into standard units
 - ▶ multiply
 - ▶ average the product

Why r Measures the **Direction** of a Linear Relation?

What is the sign of $(x_i - \bar{x})(y_i - \bar{y})$?

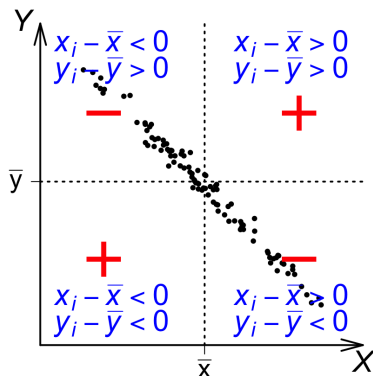


$r > 0$ as most points have
 $(x_i - \bar{x})(y_i - \bar{y}) > 0$

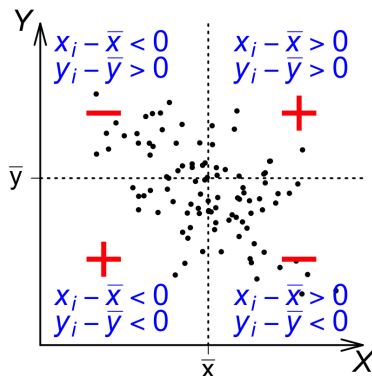


$r < 0$ as most points have
 $(x_i - \bar{x})(y_i - \bar{y}) < 0$

Why r Measures the **Strength** of a Linear Relation?



r Has a Larger Magnitude



r Has a Smaller Magnitude

Exercise: Computing the Correlation Coefficient

A TA gives a quiz with 10 questions to 5 students. The results are:

Number right (x)	Number wrong (y)
10	0
8	2
7	3
6	4
4	6

What is the correlation coefficient?

$$r = \text{Ave. of } (SU_x \times SU_y)$$

$$\bar{x} = 7, SD_x = 2, \bar{y} = 3, SD_y = 2$$

x_i	$x_i - \bar{x}$	SU_x	y_i	$y_i - \bar{y}$	SU_y	product
10	3	$3/2$	0	-3	$-3/2$	$-9/4$
8	1	$1/2$	2	-1	$-1/2$	$-1/4$
7	0	0	3	0	0	0
6	-1	$-1/2$	4	1	$1/2$	$-1/4$
4	-3	$-3/2$	6	3	$3/2$	$-9/4$

$$r = \frac{\text{average of product}}{SD_x SD_y} = \frac{1}{5} \left(\frac{-9 - 1 + 0 - 1 - 9}{4} \right) = \frac{-20}{20} = -1$$

Is there an easier way to reach this conclusion?

$$r = \text{Ave. of } (SU_x \times SU_y)$$

$$\bar{x} = 7, SD_x = 2, \bar{y} = 3, SD_y = 2$$

x_i	$x_i - \bar{x}$	SU_x	y_i	$y_i - \bar{y}$	SU_y	product
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6	-1	$-1/2$	4	1	$1/2$	$-1/4$
4	-3	$-3/2$	6	3	$3/2$	$-9/4$

$$r = \frac{\text{average of product}}{5} = \frac{1}{5} \left(\frac{-9 - 1 + 0 - 1 - 9}{4} \right) = \frac{-20}{20} = -1$$

Is there an easier way to reach this conclusion?

Yes, all points lie on a downward sloping straight line

Number wrong = 10 – Number right

Thus, r must be -1 .